

# Deep Learning — Final Project Guidelines

**Course:** Deep Learning

**Institution:** IE University

**Deliverable:** Functional MVP & Final Presentation

## 1. Project Overview

The final project for this course requires you to develop a fully functional **Minimum Viable Product (MVP)** that addresses a real-world business problem using Deep Learning. You will work in your assigned groups to build both the analytical backend and a user-facing frontend, demonstrating how advanced AI models can drive tangible business value.

## 2. Core Requirements

### Backend (The AI Model)

Your backend must be powered by a **predictive (non-generative)** Deep Learning architecture. Generative AI approaches (e.g., LLMs, GANs) are out of scope for this assignment. Depending on your business use case, your core model must utilize one of the following architectures:

- **Artificial Neural Networks (ANNs):** For tabular data, structured business metrics, or classification/regression problems.
- **Convolutional Neural Networks (CNNs):** For computer vision, image recognition, spatial data, or visual quality control.
- **Recurrent Neural Networks (RNNs / LSTMs / GRUs):** For time-series forecasting, sequential data, or natural language processing tasks.

### Frontend & Integration

- **Frontend:** You must build an interface that allows a non-technical user to interact with your model. You are fully permitted—and encouraged—to use **vibe coding** (leveraging AI-assisted tools like Cursor, vO, Bolt.new, or Streamlit) to build this quickly.
- **Integration:** A critical part of the grade is the seamless end-to-end integration. The frontend must successfully pass user inputs to your deep learning backend and display the model's predictions in real-time.

## 3. Evaluation Criteria

Your project will be graded based on three core pillars:

### Technical Depth & Execution

- Proper data preprocessing, feature engineering, and handling of datasets.

- Justification of the chosen network architecture (ANN, CNN, or RNN) and hyperparameters.
- Model evaluation using appropriate metrics (e.g., MSE, Accuracy, F1-Score, ROC-AUC) and evidence of overfitting prevention.
- Robustness of the frontend-backend integration.

### **Business Use Case & Viability**

- Clear identification of a compelling, realistic business problem.
- A well-defined value proposition: How exactly does this predictive model save time, reduce costs, or increase revenue?
- Usability of the MVP—how intuitively the frontend solves the user's pain point.

### **Presentation & Delivery**

- Clarity, professionalism, and structure of the final pitch.
- Ability to translate complex deep learning mechanics into executive-level business insights.
- Quality and smooth execution of the live MVP demo.

## **4. Final Deliverables & Submission**

You are only required to submit two assets via the campus platform:

1. **GitHub Repository Link:** Must contain clean, documented code for both the backend/model training and the frontend application, along with a brief `README.md` file on how to run the project.
2. **Final Presentation Deck:** Your presentation slides (PDF format preferred).

## **5. Presentation Day Rules**

- **Time Limit:** Exactly **15 minutes per group**. This includes your pitch, the live MVP demo, and a brief Q&A session. Time limits will be strictly enforced.
- **Participation:** **All members of the group must actively speak** during the presentation.

| Criteria                                         | Exceptional (90-100%)                                                                                                                                            | Proficient (80-89%)                                                                                                                        | Developing (70-79%)                                                                                                 | Unacceptable (<70%)                                                                                      | Weight     |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|------------|
| <b>Business Use Case &amp; Value Proposition</b> | The business problem is highly compelling and realistic. The value proposition is perfectly defined with clear impact metrics (ROI, cost reduction, efficiency). | Problem is clear and realistic. The model addresses it well, though business impact metrics could be more precise or quantified.           | The business case is vague or unrealistic. Weak connection between the model's output and actual business value.    | No clear business problem identified, or the proposed solution does not match the problem at all.        | <b>20%</b> |
| <b>Technical Depth &amp; Model Architecture</b>  | Flawless data pipeline and hyperparameter tuning. Strong architectural justification (ANN/CNN/RNN). Excellent evaluation metrics and overfitting mitigation.     | Sound model choice and training. Good evaluation, but lacks deep justification for specific hyperparameter choices or preprocessing steps. | Model runs but architecture is poorly justified. Weak evaluation metrics or clear signs of unaddressed overfitting. | Broken code, inappropriate model choice for the data, or fundamental misunderstandings of Deep Learning. | <b>25%</b> |

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| <b>MVP Integration &amp; Frontend UX</b> | Seamless frontend-backend integration. The interface (built via vibe coding) is highly intuitive, professional, and flawlessly handles real-time predictions. | Successful end-to-end integration. The interface works well, though it may have minor UX friction or lag during predictions.     | Integration is fragile or partially hardcoded . The frontend exists but struggles to communicate dynamically with the backend. | Frontend and backend are completely disconnected. The model cannot be interacted with via the interface.          | <b>25%</b> |
| <b>Presentation &amp; Team Delivery</b>  | Exceptionally professional, executive-level pitch. Seamless transitions between <b>all</b> speakers. Highly engaging visual deck.                             | Clear and professional presentation. All members spoke, though transitions may have felt slightly disjointed. Good quality deck. | Disorganized flow. Distribution of speaking time was highly unequal among members. Slides are cluttered or hard to read.       | Unprofessional delivery. Not all members spoke. Reading directly from notes/slides without engaging the audience. | <b>20%</b> |

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| <b>Live Demo &amp; Time Management</b> | Live demo worked perfectly under pressure. Presentation concluded exactly within the <b>15-minute limit</b> with time for Q&A. | Live demo was successful despite minor hiccups. Finished slightly over/under time but managed content well. | Demo failed or was replaced by static screenshots. Went significantly over time, forcing the professor to cut them off. | No demo attempted. Utter disregard for the time limit or presentation structure. | <b>10%</b> |
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